

If a Sequence is Graphical, then its Havel–Hakimi Reduction is Also Graphical

Srijan Majumder
Scholar Id : 26CS22R04

September 10, 2026

Proof

We are given a non-increasing sequence of nonnegative integers

$$S = (d_1, d_2, \dots, d_n), \quad n \geq 2, \quad d_1 \geq 1,$$

and we assume S is *graphical*, meaning some simple graph has this exact list of degrees. We must show that the shorter sequence

$$S_1 = (d_2 - 1, d_3 - 1, \dots, d_{d_1+1} - 1, d_{d_1+2}, \dots, d_n)$$

– obtained by deleting d_1 from the list and subtracting 1 from each of the next d_1 largest terms – is also graphical.

The idea, in plain words

Since S is graphical, some graph G has this degree sequence. Label its vertices $v_1, \dots, v_n$ so that $\deg(v_i) = d_i$. Vertex v_1 has degree d_1 , so it needs exactly d_1 neighbours. If we could always arrange for v_1 's neighbours to be precisely the d_1 vertices with the next-largest degrees – namely $v_2, \dots, v_{d_1+1}$ – then deleting v_1 would leave a graph on the remaining vertices where each of $v_2, \dots, v_{d_1+1}$ has lost exactly one edge (the one to v_1), and every other vertex is untouched. That leftover graph would have exactly the degree sequence S_1 . So the whole proof comes down to showing: *we can always choose a realisation of S in which v_1 's neighbours are exactly $v_2, \dots, v_{d_1+1}$.*

Step 1: Pick the best realisation

Among all graphs that realise S , choose one, G , for which v_1 is adjacent to as many of $v_2, \dots, v_{d_1+1}$ as possible. (There are only finitely many realisations, so such a "best" choice exists.)

Step 2: Show this best realisation already works

Suppose, for contradiction, that v_1 is *not* adjacent to all of $v_2, \dots, v_{d_1+1}$. We will watch what this means, and how to fix it, using an actual numerical example alongside the general argument.

A concrete "mistake" to look at. Take $S = (3, 2, 2, 2, 1)$, so $d_1 = 3$ and the target group is $\{v_2, v_3, v_4\}$. Suppose someone builds a careless realisation where v_1 connects to v_3, v_4, v_5 instead – missing v_2 and reaching out to v_5 instead. To keep every degree correct, also add edges v_2v_3 and v_2v_4 .

Every degree still matches S , so this is a valid realisation – just not the "tidy" one we want.

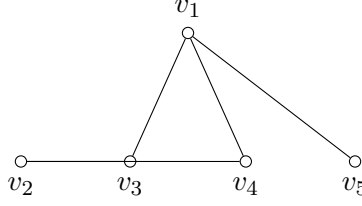


Figure 1: A careless realisation of $S = (3, 2, 2, 2, 1)$: v_1 misses v_2 and reaches v_5 instead. Degrees still match S : $\deg(v_1)=3$, $\deg(v_2)=\deg(v_3)=\deg(v_4)=2$, $\deg(v_5)=1$.

Naming the pieces. Call the missed vertex v_j and the wrongly-reached vertex v_k . Here $v_j = v_2$ and $v_k = v_5$.

- Since v_1 needs exactly $d_1 = 3$ neighbours in total, and it only found two inside the target group (v_3, v_4), it *had* to reach outside for a third – that is exactly $v_k = v_5$.
- Because the sequence is sorted largest to smallest, and $v_j = v_2$ comes earlier in the list than $v_k = v_5$, we automatically know $d_j \geq d_k$. In numbers: $\deg(v_2) = 2 \geq 1 = \deg(v_5)$.

Finding a vertex v_m to swap with. We now look for a neighbour of v_j that is *not* also a neighbour of v_k . Concretely: v_2 's neighbours are $\{v_3, v_4\}$, while v_5 's only neighbour is $\{v_1\}$. Neither v_3 nor v_4 is a neighbour of v_5 , so either one works – take $v_m = v_3$.

This always works, not just in this example, for a simple counting reason: v_j has $\deg(v_j)$ neighbours, while v_k has only $\deg(v_k) - 1$ neighbours *other than* v_1 (since one of v_k 's edges is used up connecting to v_1). Since $\deg(v_j) \geq \deg(v_k) > \deg(v_k) - 1$, vertex v_j simply has more neighbours to work with than v_k does, so at least one of v_j 's neighbours cannot also belong to v_k .

Performing the swap. Remove the two edges v_1v_k and v_jv_m (here: v_1v_5 and v_2v_3), and add the two edges v_1v_j and v_kv_m instead (here: v_1v_2 and v_5v_3).

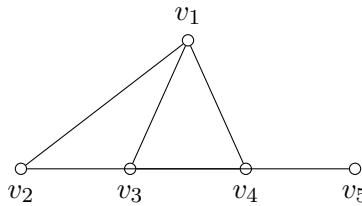


Figure 2: After the swap: v_1 now connects to exactly v_2, v_3, v_4 – the target group – and every degree in S is still satisfied.

Check the degrees after the swap: v_1 still has degree 3 (traded v_5 for v_2); v_2 still has degree 2 (traded v_3 for v_1); v_3 still has degree 2 (traded v_2 for v_5); v_4 is untouched, degree 2; v_5 still has degree 1 (traded v_1 for v_3). Nothing changed except who is connected to whom – and now v_1 hits the target group exactly.

Why this contradicts our choice of G . In general, this same swap always turns one "miss" into a "hit" without breaking any degree: v_1 loses its stray edge to v_k and gains the edge to v_j it was missing, while v_j and v_k simply trade one neighbour (v_m and v_1 respectively) without any

change to how many neighbours they have. So if G was supposed to be the realisation where v_1 already reaches as many of $v_2, \dots, v_{d_1+1}$ as possible, but it turned out to be missing one, we could always perform this swap and get an even better realisation – one contradiction. So no such "miss" can exist in the best realisation after all.

Step 3: Conclude

This means that in the best realisation G , v_1 must already be adjacent to exactly $v_2, \dots, v_{d_1+1}$. Deleting v_1 from G leaves a simple graph on the remaining $n - 1$ vertices in which each of $v_2, \dots, v_{d_1+1}$ has lost exactly one edge, and $v_{d_1+2}, \dots, v_n$ are unaffected. This leftover graph therefore has degree sequence exactly

$$S_1 = (d_2 - 1, \dots, d_{d_1+1} - 1, d_{d_1+2}, \dots, d_n),$$

proving that S_1 is graphical.

Example

Take $S = (3, 2, 2, 2, 1)$, so $n = 5$ and $d_1 = 3$. Removing d_1 and subtracting 1 from the next 3 terms gives

$$S_1 = (2 - 1, 2 - 1, 2 - 1, 1) = (1, 1, 1, 1).$$

Step 1: Build a graph realising S , with v_1 joined to the top 3

Let v_1 (degree 3) connect to v_2, v_3, v_4 (the three degree-2 vertices), and add one more edge, v_3v_4 , plus an edge v_2v_5 to finish off v_5 's single required edge.

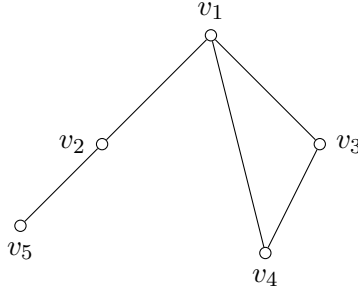


Figure 3: A realisation of $S = (3, 2, 2, 2, 1)$: $\deg(v_1) = 3$, $\deg(v_2) = \deg(v_3) = \deg(v_4) = 2$, $\deg(v_5) = 1$.

Check the degrees: v_1 has edges to v_2, v_3, v_4 (degree 3); v_2 has edges to v_1, v_5 (degree 2); v_3 has edges to v_1, v_4 (degree 2); v_4 has edges to v_1, v_3 (degree 2); v_5 has one edge, to v_2 (degree 1). This matches S exactly, and v_1 is adjacent to precisely v_2, v_3, v_4 – the top $d_1 = 3$ vertices – just as the proof requires.

Step 2: Delete v_1

Removing v_1 and all three of its edges leaves only v_3v_4 and v_2v_5 . The remaining degrees are: v_2 has degree 1, v_3 has degree 1, v_4 has degree 1, v_5 has degree 1 – exactly the sequence $(1, 1, 1, 1) = S_1$.

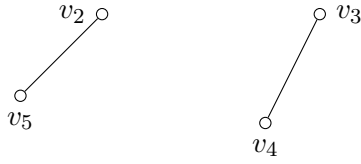


Figure 4: After deleting v_1 : two separate edges remain.

Step 3: Conclusion

Since we exhibited an actual graph with degree sequence S_1 , the sequence S_1 is graphical, confirming the theorem on this example.